

Lab: Secure a Web Server using EC2, Subnets, and Firewalls

Objective

- Launch an EC2 instance in a VPC.
- Create & attach a **Security Group** (SG).
- Create a **custom Subnet + NACL**.
- Filter traffic using SG and NACL rules.
- Verify access via browser & blocked ports.

Lab Steps

1. Create a VPC & Subnet

1. Go to **VPC** → **Your VPCs** → **Create VPC**.
 - Name: MyVPC
 - CIDR: 10.0.0.0/16
2. Under **Subnets** → **Create Subnet**
 - Name: WebSubnet
 - VPC: MyVPC
 - Availability Zone: Select any (e.g., ap-south-1a)
 - CIDR: 10.0.1.0/24
3. Enable **Auto-assign public IPv4** for the subnet.

2. Create an Internet Gateway & Attach to VPC

1. Go to **Internet Gateways** → **Create IGW**.
 - Name: MyIGW.
2. Attach it to MyVPC.
3. Update the **Route Table** of WebSubnet:
 - Add route → 0.0.0.0/0 → Target = MyIGW.

3. Create a Security Group (SG)

1. Go to **Security Groups** → **Create SG**.
 - Name: WebServer-SG
 - VPC: MyVPC
2. Add **Inbound Rules**:
 - HTTP → Port 80 → Source: Anywhere (0.0.0.0/0)
 - HTTPS → Port 443 → Source: Anywhere
 - SSH → Port 22 → Source: Your IP only (not 0.0.0.0/0 for safety)
3. Outbound Rules:
 - Default (Allow All).

4. Launch an EC2 Instance

1. Go to **EC2** → **Launch Instance**.
 - Name: MyWebServer
 - AMI: Amazon Linux 2
 - Instance Type: t2.micro (Free Tier)
 - VPC/Subnet: Select MyVPC and WebSubnet.
 - Enable Auto-assign Public IP.
 - Attach SG: WebServer-SG.
 - Key Pair: Create/download one for SSH login.

5. Install a Web Server (Apache)

SSH into the instance:

```
ssh -i mykey.pem ec2-user@<public-ip>
```

Install Apache:

```
sudo yum update -y
```

```
sudo yum install -y httpd
```

```
sudo systemctl start httpd
```

```
sudo systemctl enable httpd
```

```
echo "<h1>Hello from my EC2 Web Server</h1>" | sudo tee  
/var/www/html/index.html
```

Test:

- Open browser → `http://<EC2-Public-IP>` → Should show **Hello from my EC2 Web Server**.

6. Test Firewall Rules

- Try to access via Port 80 (HTTP) → Works
- Try via Port 22 (SSH) from another network → Blocked
- Remove HTTP rule in SG → Website becomes unreachable

7. Add NACL Rules

1. Go to **VPC** → **NACLs** → **Create NACL**.
 - Name: WebNACL
 - Attach to WebSubnet.
2. Add Rules:
 - **Inbound:** Allow 80, 443, 22
 - **Outbound:** Allow all
 - Deny everything else
3. Test again → Website should still be reachable only on allowed ports.